

Muhammad Umair Alam

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PROFILE

AI Developer specializing in **Computer Vision, LLMs, and medical imaging**, adept at **end-to-end ML development** using **PyTorch, Docker, Flask, and FastAPI**. Demonstrated success applying **cutting-edge AI solutions** in healthcare, particularly in **dental diagnostics**, through effective collaboration with cross-functional teams.

WORK EXPERIENCE

Junior AI Developer

(August, 2023 - Present)

ZIGRON, Islamabad

- Developed and implemented computer vision algorithms for diagnosing tooth diseases using OPG, bitewing and periapical X-rays, as well as mouth images.
- Utilized YOLO v8 models for segmentation of dental problems in X-ray images achieving **accuracy as high as 95%**.
- Applied post-processing techniques to implement algorithms for bone loss detection and other dental features achieving **accuracy as high as 90%**.
- Built a **scalable, Dockerized pipeline** for running **computer vision models** on dental images, built APIs on **FastAPI** for optimal performance.
- Collaborated with a team of dentists to annotate dental images using the CVAT tool, enhancing the quality of model training data.
- Developed an **AI diagnosis pipeline** for dental issues using **LangChain** and Developed a **Personalized Dental AI Chatbot** using an agentic system with **LangGraph** and **RAG (Retrieval-Augmented Generation)**, enabling intelligent and context-aware interactions for user assistance.

Software Engineer Intern

(July, 2022 - October, 2022)

Prismware Technologies

- Developed expertise in web scraping techniques using Python libraries, primarily Scrapy and Selenium
- Performed data cleaning and processing tasks using pandas, preparing datasets for analysis and model training
- Implemented CRUD operations on databases using APIs
- Contributed to the development of a chatbot using the BERT (Bidirectional Encoder Representations from Transformers) model

PROJECTS

Brain Tumor Detection using CNN	(July, 2023 - August, 2023)
Developed a Brain Tumor Detection system using binary thresholding and a CNN model, demonstrating proficient image preprocessing and deep learning skills.	

SKILLS

- Computer Vision - Pytorch - Deep Learning - Large Language Models (LLMS) - Prompt Engineering - Docker - Retrieval Augmented Generation (RAG)	- Flask - Fast Api - OpenCV - Machine Learning - CNNs - NLP - Data Science
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EDUCATION

Bachelors in Electrical Engineering	(September, 2019 - May, 2023)
<i>National University of Science and Technology</i>	Islamabad, Pakistan
<i>CGPA: 3.8</i>	
Intermediate (Pre-Engineering)	(2017 - 2019)
<i>Punjab College of Science (Blue Area Campus)</i>	Islamabad, Pakistan
<i>Passed with A*</i>	

LANGUAGES

English (full professional proficiency)	Urdu
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INTERESTS

- Computer Vision - Generative AI - Artificial Intelligence
